Tutorial 1

Problem 1. We have K objects $\{1,2,\dots,K\}$ and N boxes.
Let us denote

$$\chi_{i_1,\dots,i_m} = \begin{cases} 1, & \text{iff objects } i_1,\dots,i_m \in \text{same box} \\ 0, & \text{otherwise} \end{cases}$$

$\chi_{i_1,\dots,i_m}$ is a random variable, and number M_m of m-tuples in same boxe is

$$M_m = \sum_{i_1<\dots<i_m} \chi_{i_1,\dots,i_m}$$

Then, expected number of m-tuples in same box is

$$E(M_m) = \sum_{i_1<\dots<i_m} E(\chi_{i_1,\dots,i_m})$$

$$= \sum_{i_1<\dots<i_m} \left(0 \cdot P(\chi_{i_1,\dots,i_m}=0) + 1 \cdot P(\chi_{i_1,\dots,i_m}=1) \right)$$

$$= \sum_{i_1<\dots<i_m} \frac{1}{N^{m-1}} = \binom{K}{m} \frac{1}{N^{m-1}}$$

In our case: $K=120$, $m=3$, $N=365$

$$E(M_m) = \binom{120}{3} \frac{1}{365^2} = \frac{120\cdot 119\cdot 118}{1\cdot 2\cdot 3} \frac{1}{365^2}$$

$$\approx 2.108$$

Problem 2. a) $IDF(\text{word}) = \log_2\left(\frac{10^7}{40}\right) = \log_2(250{,}000) \approx 17.93$
≈ 18 (nearest integer)

b) $IDF(\text{word}) = \log_2\left(\frac{10^7}{10^4}\right) = \log_2(1000) \approx 9.96 \approx 10$ (nearest integer)

Problem 3. $N = 10^7$, $N_i = 320$

$$IDF(w) = \log_2\left(\frac{10^7}{320}\right) \approx 14.93$$

a)

$$TF = 1/15 \Rightarrow TF \cdot IDF = \frac{14.93}{15} \approx 0.995$$

b) $$TF = 5/15 = 1/3 \Rightarrow TF \cdot IDF \approx \frac{14.93}{3} \approx 4.97$$

Problem 4. $N = 2048$, $N_i = 128$

$$IDF = \log_2\left(\frac{2048}{128}\right) = \log_2(16) = 4$$

$$TF = 9/45 = 1/5$$

$$TF \times IDF = \frac{4}{5} = 0.8$$

Problem 5 1) Remove all columns and rows with all 0!

1)

	D_1	D_2	D_3	D_4	D_5	D_6	D_8
T_1	10	0	1	0	0	0	2
T_2	0	2	0	0	0	18	2
T_3	0	0	0	0	0	0	2
T_4	6	0	0	4	6	0	0
T_5	0	0	0	0	0	0	2
T_6	0	0	1	0	0	1	0
T_7	0	1	8	0	0	0	0
T_8	0	0	0	0	0	3	0

2) For each column (document) divide by max in this column

3) For each row multiply by $\log_2\left(\frac{7}{N_i}\right)$

(7 - number of documents
N_i - number of non-zero elements in a row)

2)

1	0	1/8	0	0	0	1
0	1	0	0	0	1	1
0	0	0	0	0	0	1
0.6	0	0	1	1	0	0
0	0	0	0	0	0	1
0	0	1/8	0	0	1/18	0
0	0.5	1	0	0	0	0
0	0	0	0	0	1/6	0

Answer:

$$\begin{pmatrix}
\log\frac{7}{3} & 0 & (\log\frac{7}{3})/8 & 0 & 0 & 0 & \log\frac{7}{3} \\
0 & \log\frac{7}{3} & 0 & 0 & 0 & \log\frac{7}{3} & \log\frac{7}{3} \\
0 & 0 & 0 & 0 & 0 & 0 & \log 7 \\
0.6\times\log\frac{7}{3} & 0 & 0 & \log\frac{7}{3} & \log\frac{7}{3} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \log 7 \\
0 & 0 & (\log\frac{7}{2})/8 & 0 & 0 & (\log\frac{7}{2})/8 & 0 \\
0 & (\log\frac{7}{2})/2 & \log\frac{7}{2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & (\log 7)/6 & 0
\end{pmatrix}$$

Problem 6. Consider $|V| \times |E|$ matrix M with

$$M(v,e) = \begin{cases} 1, & \text{if } e \text{ is an end points of } V \\ 0, & \text{otherwise} \end{cases}$$

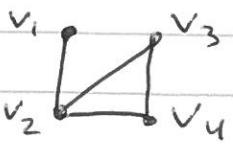
Sum of all elements in M is

a) Sum over rows $\sum_{v \in V} \deg v$

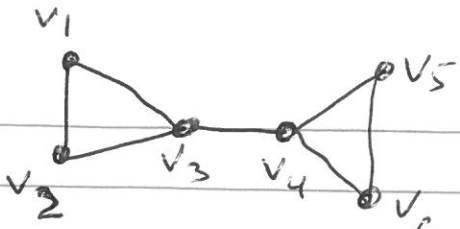
b) Sum over columns $\sum_{e \in E} 2 = 2|E|$

So, $\sum_{v \in V} \deg(v) = 2 \cdot |E|$

Problem 7



$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \end{array}$$



$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix} \end{array}$$

Problem 8

Induction on r.

1) $r = 1 \Rightarrow$ by definition of A

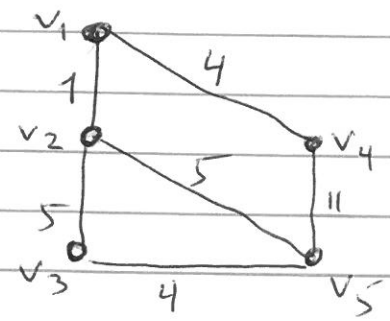
2) Suppose it is true for r. Let us denote by $p(i,j)^{(r+1)}$ the number of different walks of length $(r+1)$ from v_i to v_j. Then

$$p(i,j)^{r+1} = \sum_K p(i,K)^r \cdot p(K,j)^1 =$$

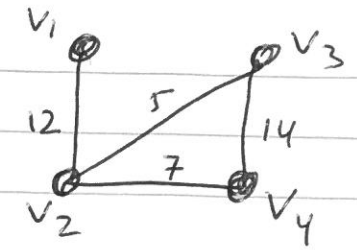
$$= \sum_K (A^r)(i,K) \cdot A(K,j)$$

$$= (A^{r+1})(i,j) \quad \text{(by definition of matrix product)}$$

Problem 9



$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_3 & v_4 & v_5 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 4 & 0 \\ 1 & 0 & 5 & 0 & 5 \\ 0 & 5 & 0 & 0 & 4 \\ 4 & 0 & 0 & 0 & 11 \\ 0 & 5 & 4 & 11 & 0 \end{pmatrix} \end{array}$$



$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{pmatrix} 0 & 12 & 0 & 0 \\ 12 & 0 & 5 & 7 \\ 0 & 5 & 0 & 14 \\ 0 & 7 & 14 & 0 \end{pmatrix} \end{array}$$

Problem 10

$X_1 = \{1, 2, 3, 4\}$ $X_2 = \{2, 3, 5, 7\}$ $X_3 = \{2, 4, 6\}$

$$sim(X_1, X_2) = \frac{2}{4+4-2} = \frac{2}{6} = 1/3$$

$$sim(X_1, X_3) = \frac{2}{4+3-2} = \frac{2}{5}$$

$$sim(X_2, X_3) = \frac{1}{4+3-1} = \frac{1}{6}$$

Problem 11

$X = \{0, 1, 2, -1, -2, -3\}$
$Y = \{-3, 3, -2, 2, -1, 1\}$

$X \cap Y = \{-2, 2, -1, 1, -3\}$

$|X \cap Y| = 5$

$$SIM(X, Y) = \frac{5}{|X| + |Y| - 5} = \frac{5}{6+6-5}$$

$$= \frac{5}{7}$$